

SIDDHANT BAPUSAHEB TANPURE

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Senior Software Engineer at Meta ◇ Distinguished Fellow, SCRS

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PROFESSIONAL SUMMARY

Technical Leader with 10+ years of industry experience specializing in AI infrastructure, distributed machine learning, trustworthy AI deployment, and large-scale data platforms. Proven track record at Meta and Microsoft building hyperscale AI technologies supporting billions of daily events. Contributor to PyTorch and Distinguished Fellow of the Soft Computing Research Society. Contributions focus on enabling scalable and resilient AI platforms capable of operating responsibly under increasingly complex global privacy, regulatory, and governance requirements, including GDPR and international data residency frameworks. Recognized for technical leadership, cross-functional influence, and contributions to the broader AI and computing community through peer review, conference service, and open-source development.

PROFESSIONAL EXPERIENCE

Ads & Monetization Infrastructure – Meta

Bellevue, Washington

Technical Leader

December 2023 – Present

- Built and scaled distributed integrity infrastructure processing more than **10B daily events** for high-throughput risk detection, safety modeling, and advertising intelligence workloads.
- Spearheaded development of privacy-aware signal frameworks and automated policy compliance systems aligned with regulatory requirements, including GDPR, and internal ad safety standards.
- Integrated high-fidelity signals into core ML ranking models and engineered observability infrastructure supporting abuse detection, reliability, and large-scale advertising intelligence initiatives associated with more than **\$400M** in long-term business impact.
- Developed high-performance ad relevance and placement systems improving contextual alignment, sensitive-surface safety, and responsible adoption for millions of businesses.
- Engineered privacy-preserving Ads Aggregate pipelines with secure data silos and compliance-first safeguards for large-scale AI and advertising systems.
- Built end-to-end ML data pipelines supplying first-party signals into ranking and LLM-assisted moderation systems to improve related ads quality, safety enforcement, and click-through performance.
- Provided technical mentorship and architectural guidance to engineers, driving adoption of responsible AI practices, scalable design patterns, and rigorous engineering standards across the team.

Microsoft Inc.

Redmond, Washington

Software Engineer II

January 2019 – December 2023

- Engineered core distributed messaging infrastructure for Microsoft Teams using C#, Cosmos DB, and distributed task framework technologies, supporting services responsible for approximately **25% of global traffic**.
- Architected and delivered Point-in-Time Restore (PITR) disaster recovery capabilities, improving platform resilience while reducing annual backup storage costs by approximately **\$1.5M**.
- Developed sovereign cloud capabilities for Microsoft Teams satisfying national security, compliance, and data residency requirements for government-grade environments.
- Designed and automated large-scale tenant deprovisioning workflows for secure data lifecycle management and global privacy compliance.
- Re-architected high-volume data pipelines for user migration and message ingestion, implementing observability with Geneva and Kusto (KQL) to maintain **99.99%** service reliability.
- Contributed to Microsoft Visual Studio developer productivity features including Hot Reload, binding diagnostics, and Property Explorer enhancements.

Rackspace

San Antonio, Texas

Software Developer – Application Engineer

July 2018 – January 2019

- Designed, built, and maintained scalable web applications for a global technical support group using Python, Flask, AngularJS, MongoDB, and Elasticsearch.
- Implemented a data-driven monitoring application to improve performance analysis for support tickets generated through chat, phone, and email channels.

OPEN SOURCE CONTRIBUTIONS

- Contributor to **PyTorch**, one of the world's leading open-source machine learning frameworks, widely used across AI research and industry.
- Worked on addressing tensor validation, CUDA and MPS backend behavior, neural-network module handling, and compiler/runtime reliability – improving the correctness and stability of foundational AI infrastructure used by researchers and practitioners worldwide.

PROFESSIONAL SERVICE & SCHOLARLY PEER REVIEW

- Reviewed **55+ manuscripts** for journals, conferences, and proceedings across artificial intelligence, distributed systems, communication systems, machine learning applications, and trustworthy computing.
- Journal and publisher reviewing activities include IEEE Access, IEEE Open Journal of the Computer Society, IEEE Transactions on Industrial Informatics, Elsevier Engineering Applications of Artificial Intelligence, Elsevier Data in Brief, Wiley International Journal of Communication Systems, Springer CCIS proceedings, and Arabian Journal for Science and Engineering.
- Conference and technical program service includes IEEE-ICCA, IEEE-ICAISSET, ICETCI, ICCIDC, CISCom, EAMCON, and DAIS.
- Served as **Technical Program Committee member** for 10+ International Conferences in Artificial Intelligence, Cybersecurity, Communications, and Applied Computing – contributing to the integrity and quality of academic and professional discourse in computing.

HONORS, FELLOWSHIPS & PROFESSIONAL RECOGNITION

- Elevated to **Senior Member of IEEE** – the highest grade of membership for which most IEEE members are eligible – recognizing significant professional achievements and technical leadership in engineering and technology.
- Conferred the title of **Distinguished Fellow** by the Soft Computing Research Society (SCRS) in recognition of outstanding professional contributions to the field of computational intelligence and applied computing.
- Invited to serve as an **international expert presenter and panelist** at the International Conference on Cybersecurity and Artificial Intelligence (ICCAI 2026), presenting on the intersection of resilient infrastructure and AI security.
- Presented as an **invited speaker** by the Fremont ACM Chapter for the May 2026 technology forum "*The Rise of Intelligent Attacks and Smarter Defenses*," delivering a talk on "*Designing Resilient, Reliable and Fault-Tolerant Infrastructure for Hyperscale AI and Cloud Platforms*."
- Invited to serve as a **Keynote Speaker and Technical Session Chair** at the 2nd International Engineering Data Analytics and Management Conference (EAMCON 2026), held at Shinawatra University, Bangkok, Thailand, contributing industry perspectives and leading technical research discussions.
- Received the **International TechXcellence Leadership Award (ITLA) 2026 – Cloud Catalyst Award** for excellence in cloud technology leadership and innovation; selected among **25 award recipients from 522 international nominations**.

EDUCATION

Arizona State University (M.S. in Software Engineering)

August 2016 – May 2018

Thesis – Analysis and improvement of Error Message System in the assembler using Think-aloud.

University of Pune (B.E. in Information Technology)

August 2010 – July 2014